Quantum network-based prediction of cancer driver genes

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Identification of cancer driver genes is fundamental for the development of targeted therapeutic interventions. The integration of mutational profiles with protein-protein-interaction (PPI) networks offers a promising avenue for their detection [Horn *et al.*, Nat. Methods **15**, 61 (2018); Nourbakhsh *et al.*, Briefings Bioinform. **25**, bbad519 (2024)], but scaling to large network datasets is computationally demanding. Quantum computing offers compact representations and potential complexity reductions. Motivated by the classical method of Gumpinger *et al.* [Bioinformatics **36**, i508 (2020)], in this work we introduce a supervised quantum framework that combines mutation scores with network topology via a state-preparation scheme we call quantum multiorder moment embedding (QMME). QMME encodes low-order statistical moments over the mutation scores of a node's immediate and second-order neighbors and encodes this information into quantum states. These states are used as inputs to a kernel-based quantum binary classifier that discriminates known driver genes from others. Simulations on an empirical PPI network demonstrate competitive performance, with a 12.6% recall gain over a classical baseline. The pipeline performs explicit quantum state preparation and requires no classical training, enabling an efficient, nearly end-to-end quantum workflow. A brief complexity analysis suggests the approach could achieve a quantum speedup in network-based cancer-gene prediction. This work underscores the potential of supervised quantum-graph-learning frameworks to advance biological discovery.

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I. INTRODUCTION

A central objective in oncology is to identify and catalog genes underlying human cancer [1–3]. While genes with high mutation rates are likely drivers, most *cancer driver genes* have few mutations. Methods that focus only on individual genes, for example, by ranking them according to how often they are mutated, cannot reliably distinguish rare cancer driver genes from irrelevant ones [2]. One explanation lies in the networked nature of cellular processes: Genes operate within pathways and complexes, so cancer may result from disruption at the pathway level rather than mutations in specific individual genes. Recent research has embraced this interaction-based perspective of cancer, integrating biological networks with statistical measures of gene-cancer associations to identify novel candidate cancer driver genes [4–8]. In these networks, nodes correspond to genes, and edges represent functional or physical interactions between them. Protein-protein-interaction (PPI) networks [9–11] are a widely used representation of gene interactions, as they integrate information from diverse data sources, tissues, and molecular processes. Previous studies combined PPI networks with gene-level cancer association scores, using clustering or network propagation methods. A common choice for these gene scores is the MutSig P value [12,13], which estimates the significance of observed mutations by considering (1) the overall mutational burden, (2) clustering of mutations within a gene's coding sequence, and (3) the functional impact of mutations. NetSig, for instance, focuses on the local neighborhood of each gene to identify cancer driver genes while correcting for node degree bias [4]. A few studies have explored supervised approaches, incorporating existing knowledge of known driver genes for the prediction of new cancer genes. Gumpinger *et al.* used Cancer Gene Census (CGC) annotations to train classifiers for discovering new cancer driver genes [14]. This method models the task as a node-level supervised classification problem in a molecular interaction network, combining network topology with mutation-score features. It achieved more than a 10% improvement in the area under the precision-recall curve (AUPRC) over baselines.

Such network-based approaches derive their strength from integrating weak mutation signals across interconnected genes rather than relying on strong evidence from individual genes. This shows the value of graph-structured representations as

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a paradigm for cancer-driver-gene discovery. However, as biological networks continue to grow in size and complexity, classical methods face increasing challenges in scalability and representational efficiency. This motivates the exploration of quantum computing as an alternative framework for representing and processing such structured data, including PPI networks. Of particular interest is quantum machine learning (QML), which has been proposed as a tool with potential for inference tasks [15–19]. Recent work has begun applying quantum computing to biological network inference problems, including link prediction tasks [20–23], demonstrating potential speedups over classical baselines, further supporting the potential of quantum approaches in this domain. Inference tasks in network biology often rely on local statistical descriptors, such as neighborhood means and low-order moments, to capture structural information in graphs [14,24]. However, quantum embedding strategies that incorporate such statistics remain unexplored. Moreover, many existing QML approaches neglect the gate complexity of state preparation, limiting their practicality as end-to-end quantum solutions [25,26]. Motivated by these gaps, we propose a quantum protocol that embeds low-order statistical moments of features in the immediate and two-hop neighborhoods in amplitude-encoded quantum states using a modular unitary construction with explicit and efficient state preparation. In this work, we introduce the *quantum multiorder moment embedding* (QMME) protocol, which encodes local graph statistics into quantum states for inference. We design an almost fully quantum pipeline for node-level binary classification of genes as cancer drivers. Classification is performed using the swap-test classifier [27], a nonparametric, distance-based method that employs a quantum kernel based on state similarity. We apply this pipeline to a PPI network, where nodes represent genes and features correspond to MutSig P values, capturing mutation significance. The entire process, from embeddings to inference, is carried out within the quantum domain, requiring no classical optimization. It relies only on minimal data-access assumptions, enabling straightforward circuit design, and involves only limited classical postprocessing, inherent in the swap-test classifier. The pipeline explicitly addresses state preparation with practical circuit construction, ensuring embeddings can be constructed with polylogarithmic qubit resources and constant-depth circuits. We validate our approach through numerical simulations on real graphs, evaluating classification performance alongside quantum resource metrics such as circuit depth and query complexity. This work establishes a proof of principle for end-to-end quantum learning on graph-structured biological data and lays the groundwork for future quantum models in complex network inference tasks.

II. PROBLEM SETUP

PPI networks model interactions between genes as nodes connected by edges, representing binding and functional associations [Fig. 1(a)]. Each gene's association with cancer can be quantified using the MutSig P value [12,13], which reflects the statistical significance of mutations considering mutational burden, clustering, and functional impact [14]. Formally, we represent a PPI network as an unweighted, undirected graph $\mathcal{G} = (V, E, \boldsymbol{x})$ with $|V| = N$ nodes, where

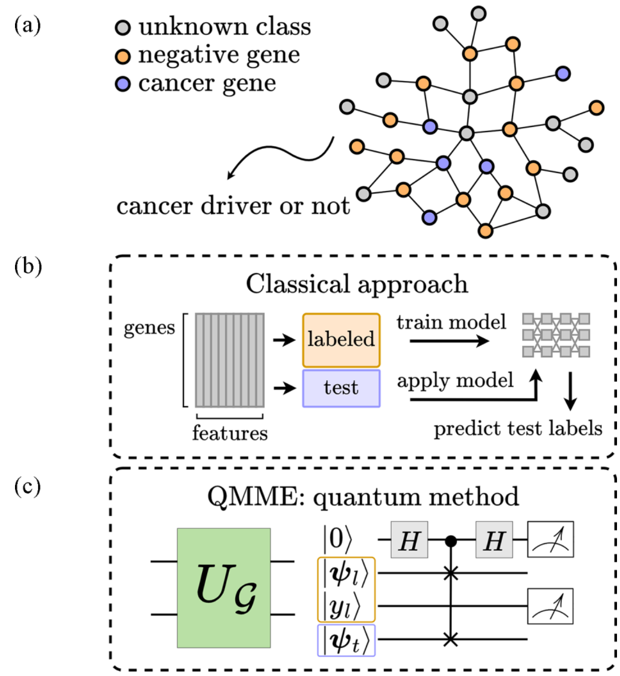

FIG. 1. Cancer-driver-gene prediction. (a) PPI network where genes are colored by class. (b) Classical machine-learning approaches: A model is trained on the labeled dataset and then used to classify the test data. (c) Our quantum approach: QMME (left) produces quantum embeddings, yielding the labeled state $|\psi_l\rangle$ and test state $|\psi_t\rangle$. The labeled state serves as a reference to classify the test state on the go via a swap test (right).

$V = [N]$ denotes the set of node indices and $E \subseteq V \times V$ is the edge set. The connectivity of $\mathcal{G}$ is described by its adjacency matrix $A \in \{0, 1\}^{N\times N}$, where $A_{vu} = 1$ indicates the presence of an edge between nodes v and u and $A_{vu} = 0$ indicates its absence. Define the immediate neighborhood of v as

$$\mathcal{N}_v^1 = \{u \in V \mid A_{vu} = 1\}. \tag{1}$$

We assume $\mathcal{N}_v^1$ is ordered by increasing node index. This allows us to define the function $r(v, \ell)$ as

$$r(v, \ell) := \mathcal{N}_v^1(\ell), \quad \ell \in \{0, \dots, |\mathcal{N}_v^1| - 1\}, \tag{2}$$

which returns the index of the ℓth neighbor of v in $\mathcal{N}_v^1$. Equivalently, $r(v, \ell)$ is the column index of the ℓth nonzero entry in row v of A. If $\ell \geqslant |\mathcal{N}_v^1|$, i.e., the function returns a flag value $*$ indicating no such neighbor exists, the two-hop neighborhood $\mathcal{N}_v^2 \subseteq V$ is defined as

$$\mathcal{N}_v^2 = \{w \in V \mid \exists u \in V, \quad A_{vu} = 1 \wedge A_{uw} = 1\}. \tag{3}$$

Each node $v \in V$ has a scalar feature value $x_v \in \mathbb{R}$, representing its MutSig P value quantifying the gene's mutation significance. The collection of all node features is denoted by the vector

$$\boldsymbol{x} = (x_1, x_2, \dots, x_N) \in \mathbb{R}^N. \tag{4}$$

The low-order origin (or raw) moments of a scalar random variable $X \sim \mathcal{P}$ are defined as $\mathbb{E}[X^i]$, where i denotes the order of the moment [28]. These moments are uncentered (i.e., calculated about the origin) and capture fundamental properties of the distribution. The first four are the mean $\mathbb{E}[X]$, the uncentered quadratic $\mathbb{E}[X^2]$, cubic $\mathbb{E}[X^3]$, and quartic $\mathbb{E}[X^4]$ moments. Low-order origin moments serve as the building blocks for the more familiar summary statistics of variance, skewness, and kurtosis, which are centered or standardized variants derived from them. Formally, we define the *origin moment embedding* as the mapping

$$\overline{\varphi}(X) = [\mathbb{E}[X], \quad \mathbb{E}[X^2], \quad \mathbb{E}[X^3], \quad \mathbb{E}[X^4]], \tag{5}$$

which collects the first four origin moments of X. Since the true distribution $\mathcal{P}$ is generally unknown, these moments are typically estimated empirically. For a finite sample $\boldsymbol{s} = [s_1, \dots, s_k]$ of X, with $s_j \in \mathbb{R}$, the corresponding *sample raw moment embedding* is given by

$$\varphi(\boldsymbol{s}) = \left[\sum_{j=1}^{k} \frac{s_j}{k}, \quad \sum_{j=1}^{k} \frac{s_j^2}{k}, \quad \sum_{j=1}^{k} \frac{s_j^3}{k}, \quad \sum_{j=1}^{k} \frac{s_j^4}{k}\right], \tag{6}$$

which returns the first four sample origin moments from the sample, thus describing its distributional shape. The hop order $c \in \{1, 2\}$ specifies the neighborhood distance from node v. Given $\boldsymbol{x}$ defined in Eq. (4),

$$\mathcal{X}_v^c = \{x_u \in \boldsymbol{x} \mid u \in \mathcal{N}_v^c\} \tag{7}$$

denotes the set of feature values of nodes at hop distance c from v, which are samples from a local distribution $\mathcal{P}_v^c$. The function $\varphi(\cdot)$ is applied to each $\mathcal{X}_v^c$, such that

$$\boldsymbol{m}_v^c := \varphi(\mathcal{X}_v^c) \in \mathbb{R}^4. \tag{8}$$

For each vertex $v \in V$, we define the *moment-feature embedding* $\boldsymbol{m}_v \in \mathbb{R}^8$ as the concatenation of $\boldsymbol{m}_v^c$ from immediate and two-hop neighborhoods:

$$\boldsymbol{m}_v = [\boldsymbol{m}_v^1, \boldsymbol{m}_v^2] \in \mathbb{R}^8. \tag{9}$$

We consider a supervised binary-classification task on a graph $\mathcal{G} = (V, E, \boldsymbol{x})$, where a subset of nodes $V_l \subseteq V$ is labeled. For these nodes, $y_v = 1$ for $v \in V_1 \subset V_l$, and $y_v = 0$ for $v \in V_0 \subset V_l$, corresponding to the positive and negative classes, respectively. This is a node-level classification task; i.e., the aim is to predict labels for individual nodes. Our goal is to predict the label $\hat{y}_v$ for any node $v \in V$ using a method that combines a *quantum node embedding* unitary operator $U_{\mathcal{G}}$, derived from the node features $\boldsymbol{x}$ and the connectivity E of $\mathcal{G}$, with a quantum binary classifier $\mathcal{C}$. This combination is used to predict whether a node v belongs to the positive or negative class. That is,

$$\mathcal{C}(U_{\mathcal{G}} |v\rangle |0\rangle^{\otimes \mathfrak{m}}) = \hat{y}_v, \tag{10}$$

where $\hat{y}_v \in \{0, 1\}$ denotes the predicted class label for node v and $\mathfrak{m}$ is the number of data qubits required to encode the node embeddings. The objective is to explicitly define $U_{\mathcal{G}}$ as the state-preparation step and design the full quantum circuit for the supervised classification of cancer driver genes. The input is a PPI network where the node features $\boldsymbol{x}$ are per-gene MutSig P values. For the labeled data, a set of positively labeled genes V_1 is obtained from CGC data, so the positive class corresponds to cancer driver genes.

III. QUANTUM MULTIORDER MOMENT EMBEDDING

We base our algorithm on having query access to the graph $\mathcal{G} = (V, E, \boldsymbol{x})$ through the adjacency list model for bounded-degree graphs [29–31]. In this input model, the graph has N vertices and bounded degree $s = 2^{\mathfrak{s}}$, i.e., at most s nonzero entries in each row of the adjacency matrix [32,33]. We assume there are no double edges in the network. To accommodate an attributed network, we extend this model by adding a feature query. Thus, the three available query operations are the following: (1) the *neighbor query*, given $v \in V$ and $\ell \in \mathbb{Z}$, returns the ℓth neighbor of v if $\ell < |\mathcal{N}_v^1|$ and $*$ otherwise, (2) the *feature query*, given $v \in V$, returns the value x_v, and (3) the *degree query*, given $v \in V$, returns the degree k_v.

In the neighbor query, if the ℓth neighbor of v does not exist (i.e., $\ell \geqslant |\mathcal{N}_v^1|$), then the flag value $*$ is returned. A quantum equivalent of this input model can be described by defining three corresponding unitary operators O_L, $\tilde{O}_X$, and $\tilde{O}_K$, acting on appropriate Hilbert spaces. Here, O_L provides access to a node's immediate neighborhood, $\tilde{O}_X$ encodes node features, and $\tilde{O}_K$ allows access to the number of c-hop neighbors. The so-called *oracles* act as

$$O_L : |v\rangle |\ell\rangle \mapsto |v\rangle |r(v, \ell)\rangle, \tag{11}$$

$$\tilde{O}_X : |v\rangle |0^{d'}\rangle \mapsto |v\rangle |\tilde{x}_v\rangle, \tag{12}$$

$$\tilde{O}_K : |v, c\rangle |0^{d'}\rangle \mapsto |v, c\rangle |\tilde{k}_v^c\rangle, \tag{13}$$

where v is the node index, ℓ is the neighbor index, $\tilde{x}_v$ is the d'-bit fixed-point representation of the node feature value $x_v \in \boldsymbol{x}$, and $\tilde{k}_v^c$ is the binary representation of the node degree $k_v^c = |\mathcal{N}_v^c|$, i.e., the number of c-hop neighbors.

Although oracle implementation on quantum hardware remains an active topic of research [34], our work assumes coherent access, allowing queries to be made in superposition, which is a standard assumption in the theoretical framework of quantum algorithms. We also assume that controlled versions of the oracles O_L, $\tilde{O}_X$, and $\tilde{O}_K$ are available, together with their inverses. Since these oracles are unitary and represented by real-valued matrices, their inverses reduce to their transposes. In terms of query complexity, neighbor and feature queries, whether classical or quantum, are considered *primitive* and are counted as $O(1)$. By contrast, a degree query can be emulated by $O(\log s)$ neighbor queries using binary search [30]. To streamline second-order neighbor access, we introduced $\tilde{O}_K$ as a generalization of the degree query, providing the size of a c-hop neighborhood for $c \in \{1, 2\}$. This oracle can be constructed from O_L with $O(\log^2 s)$ queries. The goal of the embedding circuit is to encode local graph statistics into the amplitudes of quantum states. This is done using controlled single-qubit rotations, where the rotation angles are determined by values encoded in the computational basis and retrieved from the oracles. To this end, we define two rotation operators, F_X and $F_{K^{-1}}$. Given a node feature $x_v \in \mathbb{R}$ with $|x_v| \leqslant 1$ and its binary encoding $\tilde{x}_v$, the *feature rotation operator* F_X [32] is defined as

$$F_X : |0\rangle |\tilde{x}_v\rangle \mapsto (x_v |0\rangle + \sqrt{1 - |x_v|^2} |1\rangle) |\tilde{x}_v\rangle, \tag{14}$$

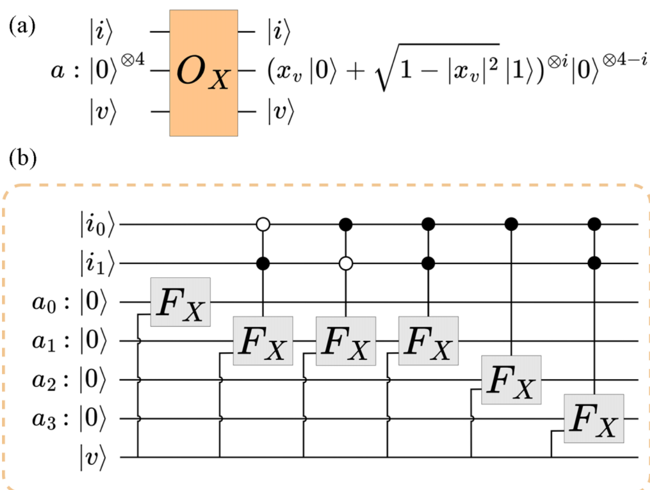

FIG. 2. Circuit of the feature oracle O_X, built from controlled rotations F_X. (a) Gate representation of O_X. (b) Detailed view: a two-qubit control register $|i\rangle$ determines how many of the four ancilla qubits undergo F_X rotations ($00 \to$ first qubit, $01 \to$ first two, $10 \to$ first three, $11 \to$ all four). A d'-qubit register is used internally but omitted for clarity.

which corresponds to a single-qubit rotation $R_y(\theta)$ with angle $\theta = 2\arcsin(x_v)$. This encodes the scalar value x_v into the amplitude of the $|0\rangle$ component. Similarly, we define the *degree-inverse rotation operator* $F_{K^{-1}}$ [35] as

$$F_{K^{-1}} : |0\rangle\,|\tilde{k}_v^c\rangle \mapsto \left(\frac{1}{k_v^c}\,|0\rangle + \sqrt{1-\left|\frac{1}{k_v^c}\right|^2}\,|1\rangle\right)|\tilde{k}_v^c\rangle. \quad (15)$$

These operators are implemented using multicontrolled R_y rotation gates [32], where the control register holds the binary input $\tilde{x}_v$ or $\tilde{k}_v^c$. The rotation angles are $\theta(\tilde{x}_v) = 2\arcsin(x_v)$ and $\theta(\tilde{k}_v^c) = 2\arcsin(1/k_v^c)$, respectively. To amplitude encode the moment-feature embedding $\boldsymbol{m}_v$ into quantum states, we introduce two operational oracles O_X and O_K^{-1}, which use the basis-encoding input oracles $\tilde{O}_X$ and $\tilde{O}_K$, along with rotation operators F_X and F_K^{-1}. The *feature oracle* O_X encodes powers of the feature value x_v conditioned on the control register $|i\rangle$. Specifically, the amplitude of the component where the ancilla register is in state $|0\rangle^{\otimes 4}$ is mapped to x_v^i, which relates to the ith-order term in the raw-moment-embedding function $\varphi(\cdot)$ from Eq. (6). The oracle O_X acts on a control register, where the integer i in the control register $|i\rangle$ dictates how many of the ancilla qubits are rotated by F_X, as follows:

$$\begin{aligned}
&|v, i\rangle\,|0\rangle^{\otimes(4+d')} \xrightarrow{\tilde{O}_X} |v, i\rangle\,|0\rangle\,|\tilde{x}_v\rangle \\
&\xrightarrow{\text{c-}F_X} |v, i\rangle\,(x_v\,|0\rangle + \sqrt{1-|x_v|^2}\,|1\rangle)^{\otimes i}|0\rangle^{\otimes(4-i)}\,|\tilde{x}_v\rangle \\
&\xrightarrow{\tilde{O}_X^T} |v, i\rangle\,(x_v\,|0\rangle + \sqrt{1-|x_v|^2}\,|1\rangle)^{\otimes i}\,|0\rangle^{\otimes(4-i+d')}. \quad (16)
\end{aligned}$$

Omitting the intermediate register $|\tilde{x}_v\rangle$, Fig. 2 shows the circuit of oracle O_X, which can be expressed as

$$O_X : |v, i\rangle|0\rangle^{\otimes 4} \mapsto |v, i\rangle\,(x_v\,|0\rangle + \sqrt{1-|x_v|^2}\,|1\rangle)^{\otimes i}|0\rangle^{\otimes(4-i)}. \quad (17)$$

The degree-inverse oracle $O_{K^{-1}}$ amplitude encodes the inverse degree scaling factors $1/k_v^c$ required to emulate the average present in the moment embedding function $\varphi(\cdot)$ included in the moment-feature embedding $\boldsymbol{m}_v$. It performs a controlled rotation that encodes the value $1/k_v^c$ as the amplitude of the $|0\rangle$ state of an ancilla as follows:

$$\begin{aligned}
&|v, c\rangle\,|0\rangle^{\otimes(1+d')} \xrightarrow{\tilde{O}_K} |v, c\rangle\,|0\rangle\,|\tilde{k}_v^c\rangle \\
&\xrightarrow{F_{K^{-1}}} |v, c\rangle\left(\frac{1}{k_v^c}\,|0\rangle + \sqrt{1-\left|\frac{1}{k_v^c}\right|^2}\,|1\rangle\right)|\tilde{k}_v^c\rangle \\
&\xrightarrow{\tilde{O}_K^T} |v, c\rangle\left(\frac{1}{k_v^c}\,|0\rangle + \sqrt{1-\left|\frac{1}{k_v^c}\right|^2}\,|1\rangle\right)|0\rangle^{\otimes d'}. \quad (18)
\end{aligned}$$

As shown in Fig. 3, the intermediate register $|\tilde{k}_v\rangle$ is omitted, and the oracle is represented as

$$O_{K^{-1}} : |v, c, 0\rangle \mapsto |v, c\rangle\left(\frac{1}{k_v^c}\,|0\rangle + \sqrt{1-\left|\frac{1}{k_v^c}\right|^2}\,|1\rangle\right). \quad (19)$$

Together with the neighbor oracle O_L and Hadamard gates, the oracles O_X and $O_{K^{-1}}$ form the core building blocks of the QMME circuit (depicted in Fig. 3). We define the QMME unitary operator $U_{\mathcal{G}}$ as

$$U_{\mathcal{G}} : |v\rangle\,|0\rangle^{\otimes \mathfrak{m}} \mapsto |v\rangle\,|\boldsymbol{\psi}_v\rangle \quad (20)$$

and refer to this mapping as the *quantum node embedding*. The second register, consisting of $\mathfrak{m}$ qubits, is called the *data register*. The total number of qubits is $\mathfrak{m} = 2\log N + 8$ (excluding work registers) and generally scales as $O(\log N)$. A schematic of the operator $U_{\mathcal{G}}$ circuit is shown in Fig. 3, and step-by-step derivation of the quantum node embedding in Eq. (20) is presented in Appendix A. The resulting state $|\boldsymbol{\psi}_v\rangle$ consists of two components:

$$|\boldsymbol{\psi}_v\rangle = |\boldsymbol{\psi}_m\rangle + |\boldsymbol{\psi}_\perp\rangle\,, \quad (21)$$

with $|\boldsymbol{\psi}_m\rangle$ denoting the *neighborhood-moment component*, which captures the quantum analog of the classical moment-feature embedding $\boldsymbol{m}_v \in \mathbb{R}^8$ defined in Eq. 9, and $|\boldsymbol{\psi}_\perp\rangle$ corresponding to an orthogonal component.

According to the circuit in Fig. 3, $|\boldsymbol{\psi}_m\rangle$ is defined as

$$\begin{aligned}
|\boldsymbol{\psi}_m\rangle &= |\boldsymbol{c} \odot \boldsymbol{m}_v\rangle\,|0\rangle^{\otimes(\mathfrak{m}-3)} && (22)\\
&= \sum_{c\in[2]} \frac{1}{2^{3/2}\,s^c} \sum_{i\in[4]} \varphi_i(\mathcal{X}_v^c)\,|c, i\rangle\,|0\rangle^{\otimes(\mathfrak{m}-3)} && (23)\\
&= \sum_{j\in[8]} c_j m_{v,j}\,|j\rangle\,|0\rangle^{\otimes(\mathfrak{m}-3)}\,, && (24)
\end{aligned}$$

where $\odot$ denotes the elementwise (Hadamard) product, $\varphi_i(\mathcal{X}_v^c)$ are the ith-order moments of the neighborhood $\mathcal{X}_v^c$, and the constant vector $\boldsymbol{c} \in \mathbb{R}^8$ is

$$\boldsymbol{c} = \frac{1}{2^{3/2}}\left[\frac{1}{s}, \frac{1}{s}, \frac{1}{s}, \frac{1}{s}, \frac{1}{s^2}, \frac{1}{s^2}, \frac{1}{s^2}, \frac{1}{s^2}\right]. \quad (25)$$

The orthogonal component $|\boldsymbol{\psi}_\perp\rangle$ of the resulting state is

$$|\boldsymbol{\psi}_\perp\rangle = \sum_{a\neq 0} |\boldsymbol{\psi}_{\perp,a}\rangle\,|a\rangle\,, \quad (26)$$

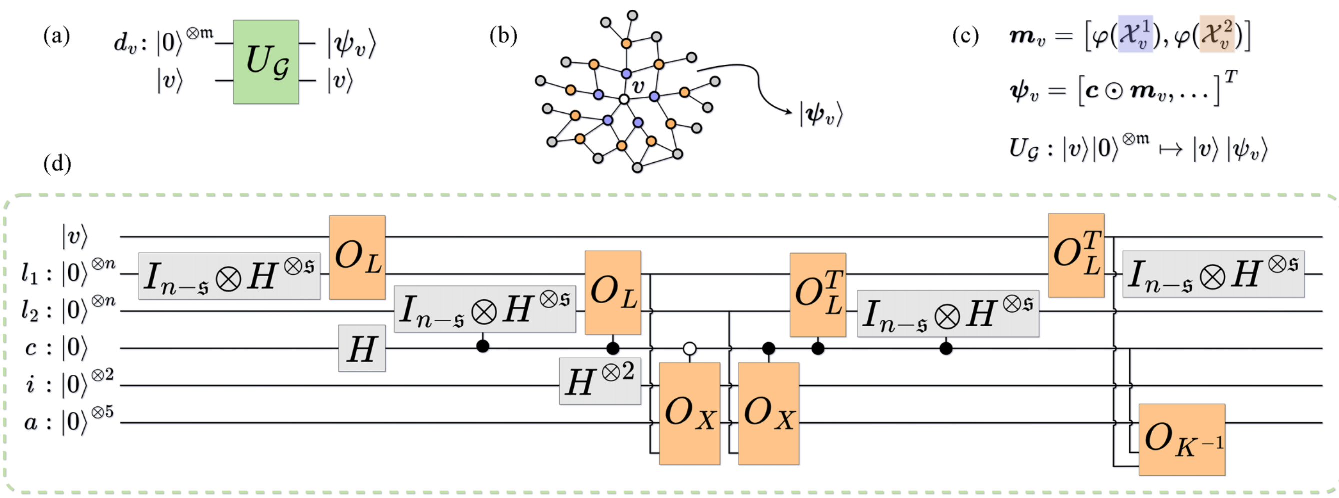

FIG. 3. QMME protocol. (a) Gate view of the unitary operator $U_{\mathcal{G}}$, which maps $|v\rangle\,|0\rangle^{\otimes\mathfrak{m}} \mapsto |v\rangle\,|\boldsymbol{\psi}_v\rangle$, amplitude-encoding low-order statistical moments of node v's neighborhood. Here, $|v\rangle$ encodes the node index $v \in V$, and the $\mathfrak{m}$-qubit data register is initialized to zero. (b) Input PPI network $\mathcal{G} = (V, E, \boldsymbol{x})$ with node features $\boldsymbol{x}$ representing mutation scores. Immediate neighbors of v are shown in purple, and second-hop neighbors are in orange. They provide the data for the embedding $|\boldsymbol{\psi}_v\rangle \in \mathbb{R}^{2^{\mathfrak{m}}}$. (c) The moment-feature embedding $\boldsymbol{m}_v$ based on the moments $\varphi(\cdot)$ over v's neighborhood is encoded as the quantum state $|\boldsymbol{\psi}_v\rangle$. (d) Circuit-level implementation of the gate $U_{\mathcal{G}}$. The $\mathfrak{m}$-qubit register is composed of subregisters for immediate and second-hop (ℓ_1, ℓ_2) neighbors, a control qubit c, a two-qubit register i indexing the ith moment, and a five-qubit ancilla register a. The circuit accesses the PPI network in (b) via the quantum oracles in Eq. (13).

with $|a\rangle$ representing computational basis states in the $(\mathfrak{m} - 3)$ data qubits and $|\boldsymbol{\psi}_{\perp,a}\rangle \in \mathbb{R}^{2^{(\mathfrak{m}-3)}}$.

As an example, take the mean of the mutation scores of the immediate neighbors of gene v, i.e., $m_{v,1} = \varphi_1(\mathcal{X}_v^1)$. As in Eq. (23), the amplitude of the state $|0_c, 0_i, 0^{(\mathfrak{m}-3)}\rangle$ is given by $\frac{\varphi_1(\mathcal{X}_v^1)}{2^{3/2}\mathfrak{s}}$. This amplitude is obtained through the following sequence of operations, illustrated in Fig. 3.

(1) Prepare an equal superposition over register ℓ_1.

(2) O_L retrieves the immediate neighbors' indices.

(3) Operator O_X encodes the neighbors' feature values into the $|0\rangle$ component of the first ancilla qubit.

(4) Uncomputing the basis register ℓ_1 via O_L^T leaves the $|0\rangle^{\otimes n}$ component amplitude proportional to the sum of neighbors' features, while others form interference patterns with alternating signs.

(5) Finally, the ancilla rotation $O_{K^{-1}}$ normalizes by the degree k_v^c, resulting in the mean of $\mathcal{X}_v^c$ encoded in the amplitude of the rotated ancillas at zero.

The sequence above describes the action of the unitary $U_{\mathcal{G}}$ for $c, i = 0$. Extending the same steps to all $c \in [2]$ and $i \in [4]$ covers both immediate and second neighbors and encodes neighborhood moments of orders 1 through 4. In general, applying $U_{\mathcal{G}}$ yields

$$\langle v, c, i, 0^{(\mathfrak{m}-3)}|\, U_{\mathcal{G}}\, |v, 0^{\mathfrak{m}}\rangle = \frac{1}{2^{3/2}\mathfrak{s}^c} \sum_{x_u \in \mathcal{X}_v^c} \frac{x_u^i}{k_v^c} = \frac{\varphi_i\big(\mathcal{X}_v^c\big)}{2^{3/2}\mathfrak{s}^c}. \tag{27}$$

IV. QUANTUM CLASSIFICATION FRAMEWORK

The quantum node embedding operator $U_{\mathcal{G}}$ maps each node in a feature-labeled network to its statistical moments, amplitude encoded within a quantum state $|\boldsymbol{\psi}_v\rangle$. Given a PPI network as input, the QMME protocol encodes local statistical moments of the MutSig P values. Depicted in Fig. 4, the quantum classification framework assumes query access to the class labels $y_{v_l} \in \{0, 1\}$ for each labeled node $v_l \in V_l$:

label query: given $v_l \in V_l$, returns the labely_{v_l},

with a quantum equivalent defined by the operator O_Y:

$$O_Y : |v_l\rangle\,|0\rangle \mapsto |v_l\rangle\,|y_{v_l}\rangle\,, \tag{28}$$

encoding the class label $y_{v_l} \in \{0, 1\}$ in a single qubit. The proposed quantum classification framework proceeds as follows: After the equal superposition over labeled nodes is prepared, the embedding unitary $U_{\mathcal{G}}$ is applied both to this register v_l

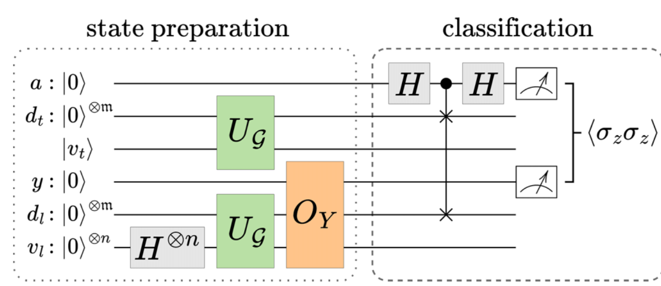

FIG. 4. Full cancer-gene prediction pipeline with the QMME protocol as part of state preparation. The first register corresponds to the ancilla qubit a, the second corresponds to the test-data qubits d_t, the third corresponds to the index qubits of the test node v_t, the fourth stores the label qubit y, the fifth stores the labeled data qubits d_l, and the sixth stores the index qubits of the labeled nodes v_l. The operator $U_{\mathcal{G}}$, together with the Hadamard gate and the oracle O_Y, prepares the quantum state, which is then processed by the classifier $\mathcal{C}$, i.e., the swap-test classifier. Node labels $\hat{y}_v \in \{0, 1\}$ are predicted via a swap-test and two-qubit measurement statistics on the ancilla and label registers. An outcome of label 1 indicates that the test gene is predicted to be a cancer driver.

and to the test-node register v_t. The class labels are loaded onto the label qubit via O_Y. Classification follows via the swap-test classifier [27,36], where the measurement statistics determine the final class prediction of the test node. The full pipeline may be described as a sequence of quantum state evolutions:

$$
\begin{aligned}
&|0\rangle\,|0\rangle^{\otimes \mathrm{m}}\,|v_t\rangle\,|0\rangle\,|0\rangle^{\otimes \mathrm{m}}\,|0\rangle^{\otimes n}\\
&\xrightarrow{H_{v_l}^{\otimes n}} \frac{1}{\sqrt{|V_l|}}\sum_{v_l\in V_l}|0\rangle\,|0\rangle^{\otimes \mathrm{m}}\,|v_t\rangle\,|0\rangle\,|0\rangle^{\otimes \mathrm{m}}\,|v_l\rangle\\
&\xrightarrow{U_{\mathcal{G}}} \frac{1}{\sqrt{|V_l|}}\sum_{v_l\in V_l}|0\rangle\,|\boldsymbol{\psi}_t\rangle\,|v_t\rangle\,|0\rangle\,|\boldsymbol{\psi}_l\rangle\,|v_l\rangle\\
&\xrightarrow{(O_Y)_{v_l,y}} \frac{1}{\sqrt{|V_l|}}\sum_{v_l\in V_l}|0\rangle\,|\boldsymbol{\psi}_t\rangle\,|v_t\rangle\,|y_l\rangle\,|\boldsymbol{\psi}_l\rangle\,|v_l\rangle\\
&\xrightarrow{H_a\cdot\text{c-swap}\cdot H_a} |\boldsymbol{\psi}_f\rangle = \frac{1}{2\sqrt{|V_l|}}\sum_{v_l\in V_l}(|0\rangle\,|\boldsymbol{\psi}_+\rangle + |1\rangle\,|\boldsymbol{\psi}_-\rangle)\,|v_l\rangle\,,
\end{aligned}
\tag{29}
$$

where $|\boldsymbol{\psi}_\pm\rangle = |\boldsymbol{\psi}_t\rangle\,(|v_t\rangle\,|y_l\rangle)\,|\boldsymbol{\psi}_l\rangle \pm |\boldsymbol{\psi}_l\rangle\,(|v_t\rangle\,|y_l\rangle)\,|\boldsymbol{\psi}_t\rangle$. The swap-test classifier computes the fidelity, i.e., the squared overlap between two states. This yields $|\langle\boldsymbol{\psi}_t|\boldsymbol{\psi}_l\rangle|^2$ for the two pure states $|\boldsymbol{\psi}_t\rangle$ and $|\boldsymbol{\psi}_l\rangle$. Fidelity serves as a similarity measure (1 if the states are identical, 0 if they are orthogonal), making it a natural distance measure in the quantum feature space. Classification thus relies directly on distances between training and test data, unifying feature mapping and classification into a single quantum procedure without requiring an intermediate classical support vector machine.

The measurement and postprocessing involve evaluating the expectation value of the two-qubit observable $\sigma_z^{(a)}\sigma_z^{(l)}$, where the superscripts indicate the ancilla and label qubit. For the final state $|\boldsymbol{\psi}_f\rangle$, $\langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle$ is given by

$$\left\langle\sigma_z^{(a)}\sigma_z^{(l)}\right\rangle = \mathrm{Tr}\left(\sigma_z^{(a)}\sigma_z^{(l)}\,|\boldsymbol{\psi}_f\rangle\,\langle\boldsymbol{\psi}_f|\right) \tag{30}$$

$$= \frac{1}{|V_l|}\sum_{v_l\in V_l}(-1)^{y_{v_l}}|\langle\boldsymbol{\psi}_t|\boldsymbol{\psi}_l\rangle|^2. \tag{31}$$

In practice, these statistics are obtained by repeating r times the two-qubit projective measurement in the σ_z basis. Using $\mathrm{Tr}(\sigma_z\,|y_{v_l}\rangle\,\langle y_{v_l}|) = 1$ for $y_{v_l} = 0$ and -1 for $y_{v_l} = 1$, the expectation value is calculated as

$$\left\langle\sigma_z^{(a)}\sigma_z^{(l)}\right\rangle = \frac{1}{r}(c_{00} - c_{01} - c_{10} + c_{11}), \tag{32}$$

where c_{al} counts measurements with the ancilla in state a and the label qubit in state l. The test data are classified as 0 if $\langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle > 0$ and 1 if negative, i.e.,

$$\hat{y}_v = \frac{1}{2}\left(1 - \mathrm{sgn}\left\langle\sigma_z^{(a)}\sigma_z^{(l)}\right\rangle\right) = \mathcal{C}(U_{\mathcal{G}}\,|v\rangle\,|0\rangle^{\otimes \mathrm{m}}), \tag{33}$$

which resembles the earlier formulation in Eq. (10). Here, $\mathcal{C}$ is the swap-test classifier, and $U_{\mathcal{G}}$ corresponds to the QMME, encoding local properties in the network $\mathcal{G}$.

V. RESULTS AND ANALYSIS

To test our quantum classification approach, we applied it to the OmniPath PPI network [37,38], accessed via the OMNIPATHR package. OmniPath integrates data from over a hundred molecular biology resources, capturing both direct physical binding, such as proteins forming complexes or modifying each other enzymatically, and functional associations, such as participation within the same metabolic pathway or biological process. This network is widely used in computational biology [39–43] and offers a framework for studying gene interactions. Nodes represent genes, and edges correspond to protein-level interactions. We extracted all pairwise gene interactions, filtered out entries with missing gene symbols, and removed duplicate edges, resulting in a network of 79 940 interactions. Each gene (node) is annotated with its $-\log_{10}$ MutSig P value [13], capturing the statistical significance of mutations in cancer. Before encoding through the quantum method, these node values are rescaled via min-max normalization to lie within [0, 1], satisfying the $|x_v| \leqslant 1$ constraint imposed by the feature rotation operator in Eq. (14) and preserving the relative mutation score of each gene. Genes without MutSig values were removed from the network. The resulting network contains 27 075 interactions and 6850 genes, forming the gene set for node classification. Positive ground-truth labels are obtained from the CGC in the COSMIC database [44]. Tier-1 genes have documented cancer-relevant activity and mutations that promote oncogenic transformation, while tier-2 genes show strong but less extensive evidence. Both tier-1 and tier-2 CGC genes are treated as positives in our experimental setup, totaling 590 known cancer genes (∼8.6% of the dataset). The remaining genes are left unlabeled. In the constructed dataset, the number of positive samples (cancer driver genes) is much smaller than the pool of negative or unlabeled samples. If left uncorrected, this class imbalance biases the classifier toward predicting negatives. To mitigate this, the labeled training set V_l is constructed for each random split by selecting 80% of the known cancer driver genes and pairing them with an equal number of randomly sampled unlabeled genes, which are provisionally assigned to the negative class [Fig. 5(c)]. This corresponds to a balanced set of 944 genes (472 positives and 472 negatives). Although some of the unlabeled genes may be undiscovered drivers, treating them as negatives provides a balanced ground truth to inform the classifier. The remaining genes form the disjoint test set V_t, and this process is repeated across 50 independent splits, with performance metrics averaged over all splits. Balancing is especially important in the quantum setting, as the first step of state preparation constructs a superposition over labeled nodes, and an imbalanced distribution would induce skewed amplitudes. For both QMME and classical moment-propagation (MoPro) embeddings, we construct a kernel matrix K from the embedding vectors and use the same distance-based classification procedure to predict labels. This ensures performance differences are attributable solely to the embeddings rather than classifier choice, providing a fair comparison. Classification labels are predicted from expectation values as in Eq. (33). The statistical significance of the differences between QMME and MoPro metrics is assessed via paired sample t tests.

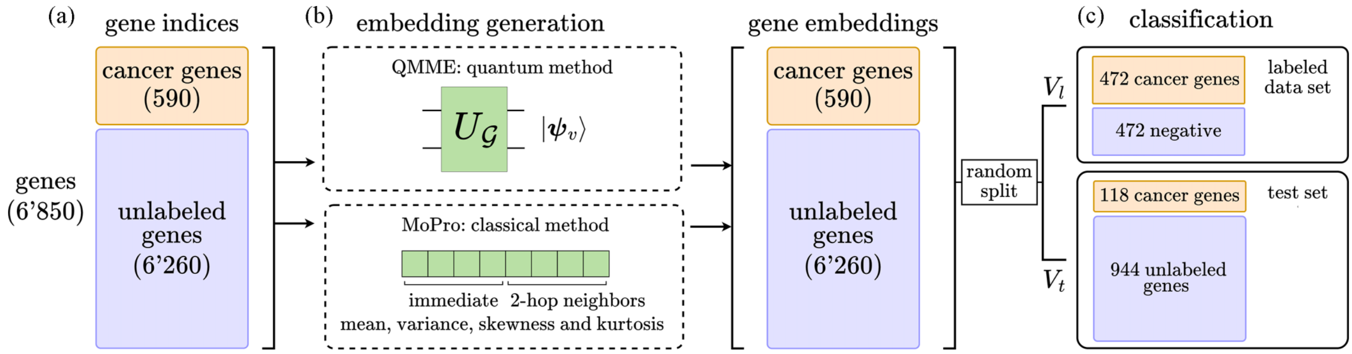

FIG. 5. Experimental setup for QMME and classical embedding comparison and performance evaluation. (a) The dataset consists of 6850 genes, of which 590 (in orange) are known cancer driver genes [44]. The remaining 6260 genes have unknown labels, as they are not part of the CGC dataset. (b) Embeddings for each gene v are obtained via two methods: the QMME protocol, and the MoPro embedding with hyperparameter $t = 1$ [14]. The QMME approach is implemented via algebraic computations on classical software. (c) In *each random split*, 80% of the positive genes are selected and combined with an equal number of unlabeled genes (assigned a negative class label) to form the labeled training set V_l. The remaining genes form the test set V_t, whose labels are used only for evaluation. For both embedding methods, kernel matrices are computed, and the expectation values in Eq. (31) are evaluated on all test genes. This process is repeated across 50 random splits, and performance metrics are averaged.

The original MoPro pipeline [14], which applies multiple classifiers with optimized hyperparameters (logistic regression, random forest, SVM, gradient boosting) and multiple propagation steps ($t > 1$), reports in its ablation study that the largest gains in performance arise from the encoding of the moments themselves, whereas additional propagation steps contribute comparatively little. In comparison, our experiments use only one-step moment embeddings and the same distance-based classifier for both QMME and MoPro. This design isolates the effect of the embeddings, avoiding unfair advantages from classifier choice or propagation, and focuses solely on the quality of the embedding representations. Nevertheless, in detail, while absolute AUPRC values in Table I are low due to the rarity of positive samples in the test set, QMME embeddings achieve a slightly higher mean AUPRC (0.0191) compared with MoPro (0.0177) and also exhibit lower variance across splits (0.0015 vs 0.0016). This indicates that QMME provides a more consistent ranking of true positives across random splits. The QMME embeddings achieve substantially higher recall (65.9%) compared with MoPro (53.3%), corresponding to a 12.6 percentage point improvement. This indicates that QMME is more effective at identifying true cancer driver genes (and rare positives) despite the low precision and overall accuracy.

While the framework is described using quantum states and operations for conceptual alignment with quantum computing principles, all quantum procedures, including the preparation of embeddings via $U_{\mathcal{G}}$, the application of the swap test, and the evaluation of expectation values, are implemented as classical simulations. Specifically, the embedding vectors and their inner products are computed algebraically on a classical computer using parallelized c++ code, without executing any operations on physical quantum hardware. This approach allows practical evaluation of the protocol while preserving the theoretical structure of the QMME and swap-test framework.

VI. COMPLEXITY ANALYSIS

We analyze the complexity of the cancer-driver-gene prediction pipeline in Table II. Let d' denote the precision (in qubits) used to encode both feature values and node degrees. The number of oracle calls required for a single application of $U_{\mathcal{G}}$ is constant. The Hadamard gates contribute a cost that scales as $O(\log s)$ within the unitary operator $U_{\mathcal{G}}$ and as $O(\log |V_l|)$ across the entire quantum pipeline. Overall, the dominant contribution to the total gate complexity arises from arithmetic operations that are part of oracle subroutines (see Appendix B for a detailed breakdown).

TABLE I. Cancer-gene classification performance (%). Comparison of quantum multiorder moment embeddings (QMMEs) and moment-propagation (MoPro) embeddings over 50 random splits. All metric differences are statistically significant except AUPRC.

Metric	QMME	MoPro	t test
Accuracy	37.2	53.1	$t = -6.829$, $p = 0.00000122$
Precision	2.1	2.3	$t = -3.486$, $p = 0.00105$
Recall	65.9	53.3	$t = 5.341$, $p = 0.000238$
F1 score	4.0	4.4	$t = -3.432$, $p = 0.00123$
AUPRC	1.9 ± 0.2	1.8 ± 0.2	

TABLE II. Complexity analysis of the full cancer-driver-gene prediction pipeline, including state preparation (with the QMME protocol) and classification, reported in terms of gate count, qubit requirements, circuit depth, and query complexity (calls to primitive oracles).

Metric	Complexity
Gate complexity	$O(\text{polylog}\, N + d')$
Qubit count	$O(\log N + d')$
Circuit depth	$O(\log N)$
Query complexity	$O(\log^2 s)$

Circuit depth is the number of sequential gate layers, i.e., the computation length, and measures time complexity. The number of qubits is the size of the quantum register, which measures space (memory) complexity. For the embedding circuit $U_{\mathcal{G}}$, time complexity is independent of the network size, and space complexity scales polylogarithmically. Considering the entire classification pipeline, the swap-test classifier, which relies on controlled-swap operations to estimate quantum state overlap, introduces a logarithmic dependence on the feature dimension $2^{\mathfrak{m}}$ in its circuit depth. Since $\mathfrak{m} = O(\log N)$, this results in an overall logarithmic dependence on the network size N.

VII. DISCUSSION AND CONCLUSIONS

In this work, we introduced an almost fully quantum pipeline for cancer-driver-gene prediction based on a swap-test classifier, requiring no classical optimization loops. All stages of encoding, embedding, and inference are performed within the quantum domain, requiring only minimal data-access assumptions and limited classical postprocessing. Our approach demonstrates polylogarithmic resource scaling together with competitive performance on protein-protein-interaction network data. The pipeline requires $O(\log N + d')$ qubits and $O(\text{polylog}\, N)$ primitive gates, with an additional logarithmic dependence on the sparsity s in terms of query complexity. The proposed quantum multiorder moment embedding is a modular, extensible, and general-purpose architecture for encoding local graph statistics into quantum states. While we focused on binary classification of genes as cancer drivers using PPI networks (with nodes representing genes and MutSig P values as features), QMME is broadly applicable to node-level classification tasks on feature-labeled graphs across domains [24]. The current setup formulation assumes unweighted and undirected edges, and generalization depends on the informativeness of local neighborhood statistics in those networks. Additional feature preprocessing or more sophisticated classification algorithms can be incorporated with minimal modification, allowing easy adaptation to different network types or learning objectives. Future directions include extending QMME to capture higher-order moments; supporting directed, weighted, and multilayered graphs; enabling multiclass classification; and implementing resource-efficient variants suitable for near-term quantum hardware.

VIII. LIMITATIONS AND PATH TO QUANTUM IMPLEMENTATION

While the proposed pipeline exhibits favorable polylogarithmic scaling in theory, its practical implementation on current quantum hardware remains challenging [45]. Near-term (NISQ) devices are limited in qubit number, coherence times, and gate fidelity, restricting the feasible circuit depth and system size [26,46]. The framework assumes efficient oracle access to graph structure and node features and does not account for the cost of loading these classical data into quantum states [47,48], which may affect practical speedups. Moreover, expectation values in the swap-test classifier require repeated measurements, with the number of shots scaling as $O(1/\epsilon^2)$ for estimation error ϵ. Finally, the classification framework is described for predicting the label of a single test node, which serves as a simplification for establishing a proof of concept.

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B.C. and D.M. conceived the initial concept of the project. A.W. made substantial contributions to the machine-learning components. D.M. assisted with the complexity analysis of the quantum circuit. P.M. developed the framework, performed all calculations and simulations, and developed the main tools, in close collaboration with B.C. and A.W. P.M. drafted the manuscript, and B.C., A.W., and D.M. provided critical review and revisions.

The authors declare no competing interests.

DATA AVAILABILITY

The data that support the findings of this article are openly available [49], embargo periods may apply.

APPENDIX A: QUANTUM NODE EMBEDDING STEP-BY-STEP DERIVATION

Let U_r denote the operator U acting on register r, with identity on all other registers. The notation c-U_r represents the controlled application of U_r, with the control qubit in register c in state $|1\rangle$. Similarly, $\mathrm{c}^{(0)}$-U_r denotes control on the $|0\rangle$ state of c. Below, we describe the evolution of the quantum state through the quantum multi-order moment embedding (QMME) protocol defined in Eq. (20), step by step, including a brief description and gate complexity estimate for each transformation (e.g., oracle applications and other unitary operations).

Start with the initial state. The node v is encoded, and all other qubits are initialized to zeros,

$|v, 0^{\mathfrak{m}}\rangle$, where $\mathfrak{m} = 2\log N + 8$, omitting work qubits. Put the register l_1 into an equal superposition over all possible neighbor indices $\ell_1 \in [s]$; counts as $O(\log s)$ in gate complexity:

$$\xrightarrow{I_{n-\mathfrak{s}} \otimes H_{l_1}^{\otimes \mathfrak{s}}} \frac{1}{\sqrt{s}} |v, 0^3\rangle \sum_{\ell_1 \in [s]} \left|\ell_1, 0^{(n+5)}\right\rangle.$$

Map each ℓ_1 in superposition to the actual index of the corresponding neighbor of v, which requires $O(\text{polylog}\, N)$ elementary gates:

$$\xrightarrow{(O_L)_{v,l_1}} \frac{1}{\sqrt{s}} |v, 0^3\rangle \sum_{\ell_1 \in [s]} \left|r(v, \ell_1), 0^{(n+5)}\right\rangle.$$

Put the control qubit c into superposition (for immediate and second-order neighborhoods), which requires a single elementary gate:

$$\xrightarrow{H_c} \frac{1}{\sqrt{2s}} |v\rangle \sum_{c\in[2]} \sum_{\ell_1\in[s]} \left|c, 0^2, r(v,\ell_1), 0^{(n+5)}\right\rangle .$$

If $c = 1$, create a superposition over all second-neighbor indices ℓ_2, capturing the two-hop neighborhood ($O(\log s)$ gates):

$$\xrightarrow{I_{n-\mathfrak{s}}\otimes \text{c-}H_{\ell_2}^{\otimes \mathfrak{s}}} \frac{1}{\sqrt{2s}} |v\rangle \sum_{\ell_1\in[s]} \left(\left|0_c, 0^2, r(v,\ell_1), 0^{(n+5)}\right\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2\in[s]} |1_c, 0^2, r(v,\ell_1), \ell_2, 0^5\rangle \right).$$

If $c = 1$, retrieve the indexes of second-order neighbors of v, which requires $O(\text{polylog}\, N)$ gates:

$$\xrightarrow{\text{c-}(O_L)_{\ell_1,\ell_2}} \frac{1}{\sqrt{2s}} |v\rangle \sum_{\ell_1\in[s]} \left(\left|0_c, 0^2, r(v,\ell_1), 0^{(n+5)}\right\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2\in[s]} \left|1_c, 0^2, r(v,\ell_1), r\big(\mathcal{N}_v^1(\ell_1), \ell_2\big), 0^5\right\rangle \right).$$

Create an equal superposition over the four moment indices i, which requires $O(1)$ gates:

$$\xrightarrow{H_i^{\otimes 2}} \frac{1}{2^{3/2}\sqrt{s}} |v\rangle \sum_{\ell_1\in[s]} \sum_{i\in[4]} \left(\left|0_c, i, r(v,\ell_1), 0^{(n+5)}\right\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2\in[s]} \left|1_c, i, r(v,\ell_1), r\big(\mathcal{N}_v^1(\ell_1), \ell_2\big), 0^5\right\rangle \right).$$

If $c = 0$, amplitude encode the ith power of the immediate neighbors' features; gate cost is $O(\text{polylog}\, N + d')$:

$$\xrightarrow{\text{c}^{(0)}\text{-}(O_X)_{i,a,l_1}} \frac{1}{2^{3/2}\sqrt{s}} |v\rangle \sum_{\ell_1,i} \left(\left[x^i_{r(v,\ell_1)} \left|0_c, i, r(v,\ell_1), 0^{(n+4)}\right\rangle + |\perp\rangle\right] |0\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2\in[s]} \left|1_c, i, r(v,\ell_1), r\big(\mathcal{N}_v^1(\ell_1), \ell_2\big), 0^5\right\rangle \right).$$

If $c = 1$, amplitude encode the ith power of the second-hop neighbors' features; $O(\text{polylog}\, N + d')$ is the gate complexity:

$$\xrightarrow{\text{c-}(O_X)_{i,a,\ell_2}} \frac{|v\rangle}{2^{3/2}\sqrt{s}} \sum_{\ell_1,i} \left(x^i_{r(v,\ell_1)} \left|0_c, i, r(v,\ell_1), 0^{(n+5)}\right\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2\in[s]} x^i_{r(\mathcal{N}_v^1(\ell_1),\ell_2)} \left|1_c, i, r(v,\ell_1), r\big(\mathcal{N}_v^1(\ell_1), \ell_2\big), 0^5\right\rangle \right) + |v\rangle\, |\perp\rangle\, |0\rangle .$$

If $c = 1$, rewrite the second-hop neighbor label $r(\mathcal{N}_v^1(\ell_1), \ell_2)$ back into its index ℓ_2; gate cost is $O(\text{polylog}\, N)$:

$$\xrightarrow{\text{c-}(O_L^T)_{\ell_1,\ell_2}} \frac{1}{2^{3/2}\sqrt{s}} |v\rangle \sum_{\ell_1,i} \left(x^i_{r(v,\ell_1)} \left|0_c, i, r(v,\ell_1), 0^{(n+5)}\right\rangle + \frac{1}{\sqrt{s}} \sum_{\ell_2\in[s]} x^i_{r(\mathcal{N}_v^1(\ell_1),\ell_2)} |1_c, i, r(v,\ell_1), \ell_2, 0^5\rangle \right) + |v\rangle\, |\perp\rangle\, |0\rangle .$$

If $c = 1$, uncompute the ℓ_2 superposition, resetting the index register (sum second neighbors' features); this requires $O(\log s)$ gates:

$$\xrightarrow{I_{n-\mathfrak{s}}\otimes \text{c-}H_{\ell_2}^{\otimes \mathfrak{s}}} \frac{1}{2^{3/2}\sqrt{s}} |v\rangle \sum_{\ell_1,i} \left(x^i_{r(v,\ell_1)} |0_c, i, r(v,\ell_1)\rangle + \frac{1}{s} \sum_{x'\in\mathcal{X}^1_{r(v,\ell_1)}} x'^i\, |1_c, i, r(v,\ell_1)\rangle \right) \left|0^{(n+5)}\right\rangle + |v\rangle\, |\perp\rangle\, |0\rangle .$$

Rewrite the immediate-neighbor lookup $r(v, \ell_1)$ back to its index ℓ_1, which requires $O(\text{polylog}\, N)$ gates:

$$\xrightarrow{(O_L^T)_{v,l_1}} \frac{1}{2^{3/2}\sqrt{s}} |v\rangle \sum_{\ell_1,i} \left(x^i_{r(v,\ell_1)} |0_c, i, \ell_1\rangle + \frac{1}{s} \sum_{x'\in\mathcal{X}^1_{r(v,\ell_1)}} x'^i\, |1_c, i, \ell_1\rangle \right) \left|0^{(n+5)}\right\rangle + |v\rangle\, |\perp\rangle\, |0\rangle .$$

Uncompute the ℓ_1 superposition, resetting the index register (sum immediate neighbors' features), which requires $O(\log s)$ gates:

$$\xrightarrow{I_{n-\mathfrak{s}}\otimes H_{l_1}^{\otimes \mathfrak{s}}} \frac{1}{2^{3/2}s} |v\rangle \sum_{i\in[4]} \left(\sum_{x'\in\mathcal{X}^1_{r(v,\ell_1)}} x'^i\, |0_c\rangle + \frac{1}{s} \sum_{x''\in\mathcal{X}^2_v} x''^i\, |1_c\rangle \right) \left|i, 0^{(2n+5)}\right\rangle + |v\rangle\, |\perp\rangle\, |0\rangle .$$

TABLE III. Quantum gate complexity and queries for a single pipeline run. Query complexity counts the number of calls to the primitive oracles O_L and $\tilde{O}_X$.

Operator	Description	Gate complexity	Qubit count	Query complexity
H	elementary gate	1	1	
O_L	input model	$O(\operatorname{polylog} N)$	$O(\log N)$	$O(1)$
$\tilde{O}_X$	input model	$O(\operatorname{polylog} N + d')$	$O(\log N + d')$	$O(1)$
$\tilde{O}_K$	derived from O_L	$O(\operatorname{polylog} N + d')$	$O(\log N + d')$	$O(\log^2 s)$
F_X	type of R_y	$O(d')$	$O(d')$	
$F_{K^{-1}}$	type of R_y (inverse)	$O(d')$	$O(d')$	
O_X	$1\tilde{O}_X \cdot 5\,\text{c-}F_X \cdot 1\tilde{O}_X^T$	$O(\operatorname{polylog} N + d')$	$O(\log N + d')$	$O(1)$
$O_{K^{-1}}$	$1\tilde{O}_K \cdot 1F_{K^{-1}} \cdot 1\tilde{O}_K^T$	$O(\operatorname{polylog} N + d')$	$O(\log N + d')$	$O(\log^2 s)$
QMME	embedding protocol	$O(\operatorname{polylog} N + d')$	$O(\log N + d')$	$O(\log^2 s)$
Cancer-driver prediction pipeline	see Fig. 4	$O(\operatorname{polylog} N + d')$	$O(\log N + d')$	$O(\log^2 s)$

Normalize by v's degree; amplitudes are averaged neighbor statistics rather than sums, and gate complexity is $O(\operatorname{polylog} N + d')$:

$$\begin{aligned}
&\xrightarrow{(O_{K^{-1}})_{v,k}} \frac{1}{2^{3/2}s}\,|v\rangle \sum_{i\in[4]} \left(\sum_{x'\in\mathcal{X}^1_{r(v,\ell_1)}} x'^i\,|0_c\rangle + \frac{1}{s}\sum_{x''\in\mathcal{X}^2_v} x''^i\,|1_c\rangle \right) \frac{1}{k^c_v}\left|i, 0^{(2n+5)}\right\rangle + |v\rangle\,|\perp\rangle \\
&= |v\rangle \left(\sum_{i,c} \frac{1}{2^{3/2}s^c} \sum_{x_u\in\mathcal{X}^c_v} \frac{x^i_u}{k^c_v}\left|c, i, 0^{(2n+5)}\right\rangle + |\perp\rangle \right) \\
&= |v\rangle \left(\sum_{i,c} \frac{1}{2^{3/2}s^c}\varphi(\mathcal{X}^c_v)\left|c, i, 0^{(2n+5)}\right\rangle + |\perp\rangle \right) \\
&= |v\rangle \left(\sum_{j\in[8]} c_j m_{v,j}\left|j, 0^{(2n+5)}\right\rangle + |\perp\rangle \right),
\end{aligned}$$

with $\boldsymbol{c}$ defined as in Eq. (25).

APPENDIX B: COMPLEXITY ANALYSIS IN DETAIL

We can construct a quantum circuit to implement the primitive oracles O_L and $\tilde{O}_X$ with, respectively, $O(\operatorname{polylog} N)$ and $O(\operatorname{polylog} N + d')$ elementary gates. O_L performs a neighbor lookup, while $\tilde{O}_X$ encodes the d'-bit feature vector of a node into a binary register. Thus, the cost of implementing neighborhood and feature access oracles is polylogarithmic in the graph size, with an additional linear factor in the precision d' for $\tilde{O}_X$. Controlled versions of these oracles, as well as their inverses, inherit the same asymptotic cost. We assume costs consistent with standard data structures for this problem class, as in Remark 3 from Tong *et al.* [35,50,51]:

> If we consider a diagonal matrix $D \in \mathbb{R}^{N\times N}$ and assume that each diagonal entry is accessible via an oracle
>
> $$O_D\,|i\rangle\,|0^l\rangle = |i\rangle\,|D_{ii}\rangle\,, \quad i\in[N],$$
>
> where $|D_{ii}\rangle$ is the l-bit binary representation of D_{ii}, then each diagonal entry of D can be efficiently computed using a classical Boolean circuit of size $\mathcal{O}(\operatorname{polylog} N)$ with $m = \mathcal{O}(\operatorname{polylog} N)$ ancilla bits. In this case, one can construct a quantum circuit with $\mathcal{O}(\operatorname{polylog} N + l)$ gates and $\mathcal{O}(\operatorname{polylog} N + l)$ ancilla qubits to implement this oracle O_D.

The rotation operators are simply controlled versions of single-qubit $R_y(\theta)$ rotations, with the angle computed from the d' qubits in the control register. Using simple fixed-point arithmetic, such operators scale as $O(d')$.

Recall that gene labels are predicted using a swap test and two-qubit measurement statistics. In terms of gate count, the controlled-swap gate, which exchanges the training data and the test data, requires only $O(\log|V_l|)$ elementary gates to implement (Supplemental Material of [27]). The two-qubit measurements are repeated r times to estimate the expectation value $\langle\sigma_z^{(a)}\sigma_z^{(l)}\rangle$, as defined in Eq. (32). As such, the overall cost for this prediction scales linearly with r, while the per-run complexities are those listed in Table III. The statistical error of the estimate decreases with the number of repetitions r as $\sqrt{r}$, following standard sampling statistics. As such, to get an estimate with error at most ϵ, the number of repetitions must scale as $r = O(1/\epsilon^2)$.

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